



Zahran Toric Tool

USER MANUAL

VERSION

1.0.0

PLATFORM

iOS / Android

DATE

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A clinical-grade mobile application for toric intraocular lens planning, IOL alignment, and astigmatism outcome analysis — designed for ophthalmic surgeons.

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CHAPTER 1

Introduction & Overview

What the Zahran Toric Tool does and how to use it safely.

1.1 Purpose of the Application

The **Zahran Toric Tool** is a mobile clinical decision-support tool designed to assist ophthalmic surgeons in planning and executing toric intraocular lens (IOL) implantation. It combines pre-operative biometric calculation, intraoperative axis alignment, and post-operative outcome analysis in a single cohesive workflow.

The application follows established ophthalmic conventions including TABO axis notation (0°–180°) and the double-angle vector method (DAVM) for all astigmatism calculations.

1.2 Key Features



Toric IOL Calculator

Calculates residual astigmatism, IOL cylinder power, and optimal target axis from corneal biometry.



Premium Camera Capture

Instrument-gauge dial with TABO degree markings, live tilt sensor, and centre level readout.



IOL Alignment Overlay

Real-time reference and target axis lines overlaid on the captured eye photo.



Landmark Annotation

Tap-to-place vessel reference markers on the eye photo for pre-operative marking.



Alpins Vector Analysis

Full post-operative outcome analysis: TIA, SIA, DV, CI, ME, AE, and IS metrics with double-angle plot.



SIA Nomogram Tracker

Builds a personal SIA nomogram from your cases to personalise future calculations.



PCA Correction

Barrett-style Posterior Corneal Astigmatism correction for more accurate toric planning.



Post-Refractive Mode

Adapted planning for eyes with prior LASIK, PRK, or RK surgery.

1.3 Medical Disclaimer

⚠ Important Medical Notice

This application is intended as a **clinical decision-support tool only**. It does not replace the professional judgment of a qualified ophthalmic surgeon. All calculations must be reviewed and verified by the treating surgeon before any clinical action is taken. The developer assumes no liability for outcomes resulting from use of this application. Always cross-reference results with your IOL manufacturer's nomogram and current clinical guidelines.

CHAPTER 2

Getting Started

Installation, permissions, and your first case.

2.1 Installation & Permissions

The app requires the following device permissions:

PERMISSION	PURPOSE	REQUIRED?
Camera	Capture anterior segment eye photos for alignment	REQUIRED
Photo Library	Save captured eye images to device gallery (optional)	OPTIONAL
Motion Sensors	Device tilt detection for leveling the camera	REQUIRED

2.2 Navigation Overview

SCREEN HIERARCHY

- **Patient List** → Patient Detail → Toric Calculator → (Camera → Alignment)
- **Patient Detail** → Alpina Analysis
- **Patient Detail** → Landmark Annotation
- **Patient List** (header) → SIA Nomogram
- **Patient List** (header) → Settings

2.3 Workflow Summary

- 1 **Create a patient record** — enter name, DOB, and any notes.
- 2 **Add an eye record** — enter keratometry (K1/K2), IOL power, and cylinder data.
- 3 **Run the Toric Calculator** — review recommended IOL cylinder and axis.
- 4 **Capture the eye photo** — use the premium camera gauge to align and capture.

- 5 **Align the IOL** — use the alignment overlay in the operating room.
- 6 **Post-operatively** — enter actual refraction and run Alpina Analysis.
- 7 **Log the case to your SIA Nomogram** — build your personalised tracker over time.

CHAPTER 3

Patient Management

Creating and organising your patient database.

3.1 Patient List Screen

The Patient List is the home screen. All patients are stored locally on your device. The list shows name, date of birth, and the number of eye records.

- Tap **+ New Patient** to add a record.
- Tap any patient row to open their detail page.
- Tap the **SIA** button (top-right) to open the SIA Nomogram.
- Tap the ⚙ gear icon to open Settings.

3.2 Adding / Editing a Patient

FIELD	DESCRIPTION
Full Name	Patient's full legal name
Date of Birth	DD/MM/YYYY format
Notes	Free-text clinical notes (allergies, comorbidities, etc.)

3.3 Eye Records

FIELD	UNIT	DESCRIPTION
Eye	—	OD (Right) or OS (Left)
K1 Power	D	Flat corneal meridian power (lower value)
K1 Axis	°	TABO axis of flat meridian (0–180°)
K2 Power	D	Steep corneal meridian power (higher value)
IOL Power	D	Spherical IOL power planned

FIELD	UNIT	DESCRIPTION
IOL Cylinder	D	Toric IOL cylinder power at IOL plane
Target Axis	°	Planned axis for IOL alignment (TABO)
SIA / SIA Axis	D / °	Expected surgically-induced astigmatism magnitude and axis
Post-Refractive	Toggle	Enable if eye had prior refractive surgery
Procedure Type	LASIK/PRK/RK	Type of prior refractive procedure

Tip

K1 must be the flattest meridian (lower power) and K2 the steepest (higher power). The calculator handles the vector arithmetic automatically.

CHAPTER 4

Toric Calculator

Computing optimal IOL cylinder power and alignment axis.

4.1 Entering Biometric Data

Navigate to the Toric Calculator from the Patient Detail screen. Biometric data already saved in the eye record is pre-populated automatically. The calculator requires K1/K2 powers and axes, IOL cylinder, SIA, SIA axis, and target residual astigmatism.

4.2 PCA Correction (Barrett Posterior Corneal Astigmatism)

The **PCA Correction toggle** applies the Barrett method, accounting for the ~0.3 D against-the-rule contribution of the posterior corneal surface not captured by anterior keratometry.

When to enable PCA Correction

Enable for all patients unless you have direct posterior corneal measurements (e.g. Pentacam TCRP data).

```
// Double-angle vector components of anterior astigmatism
Ax = (K1 - K2) · cos(2 × K1_axis_rad)
Ay = (K1 - K2) · sin(2 × K1_axis_rad)

// Add 0.3 D ATR posterior contribution
Tx = Ax + 0.3 | Ty = Ay

// Convert back to magnitude / axis
adjustedMag = √(Tx² + Ty²)
adjustedAxis = ½ · atan2(Ty, Tx) × (180/π) [mod 180°]
```

4.3 Post-Refractive Surgery Mode

When the eye has had prior corneal refractive surgery, enable the **Post-Refractive toggle** in the eye record and select the procedure type (LASIK, PRK, or RK). A warning banner is displayed in the calculator prompting use of dedicated post-refractive formulas (e.g. Barrett True-K, Haigis-L).

Post-Refractive Cases

Standard toric formulas are less reliable after corneal refractive surgery. Always cross-reference with a dedicated post-refractive IOL calculator.

4.4 Results & Surgical Plan Export

After tapping **Calculate**, the results panel shows: recommended target axis, expected residual astigmatism, correction index estimate, and a rotation effect chart (residual vs. IOL rotation 0°–30°).

Tap **Export Plan** to generate a PDF surgical plan with patient details, biometry, calculated axes, and the rotation effect chart. Share via any installed share sheet (email, AirDrop, etc.).

CHAPTER 5

Camera & Eye Capture

Capturing a clinical-quality anterior segment image.

5.1 Instrument Gauge Overlay

ELEMENT	COLOUR	DESCRIPTION
Outer accent rings (×3)	■ Gold	Decorative concentric borders at R = 142, 134, 127 px
Main guide ring	■ Gold	Primary 120 px radius ring — align the limbus to this
Tick marks — major	Gold	Every 30°, tall and bold. Labelled with degree value.
Tick marks — medium	Semi-gold	Every 10°, medium length
Tick marks — minor	Faint gold	Every 5°, short
Axis labels	Gold	TABO degrees: 0°, 30°, 60°, 90°, 120°, 150°, 180°
Inner pupil guide	■ Cyan dashed	54 px radius ring — align the pupil margin to this
Crosshairs	White (30% opacity)	Horizontal and vertical centre guides
Centre readout	Gold / Green	Shows live tilt angle; turns green when within $\pm 3^\circ$

5.2 Leveling the Device

The app reads the device's **roll angle** from the gyroscope at 10 Hz. The centre circle turns **green** and the shutter button glows green when device tilt is $\leq \pm 3^\circ$.

5.3 Capture Steps

- 1 Open the Camera screen from Patient Detail.
- 2 Align the limbus to the main gold ring and the pupil to the cyan inner ring.
- 3 Wait for the centre readout to show **LEVEL** (tilt $\leq \pm 3^\circ$).

4

Tap the shutter button (glows green when level). The app saves the image and proceeds to the Alignment screen.

CHAPTER 6

IOL Alignment Screen

Using the alignment overlay in the operating room.

6.1 Overlay Elements

ELEMENT	COLOUR	MEANING
Reference axis line	 Blue	Pre-operative corneal reference meridian (K1 axis)
Target axis line	 Green	Desired final IOL axis — align the IOL markings to this
Current axis line	 Gold	Current adjusted rotation angle
Limbal ring	 Cyan	Guide circle for limbal registration
Gold tick marks	 Gold	Degree markers every 10° around the limbal ring
Landmark dots	Colour-coded	Pre-marked vessel reference points

6.2 Manual Axis Adjustment

Use the **+/-** degree buttons to manually rotate the overlay axes in 1° increments to account for cyclorotation or slight image tilt.

6.3 Landmark Annotations

If vessel landmarks were saved (see Chapter 7), they appear as coloured ring-and-dot markers confirming the correct rotational orientation of the eye.

- **Recapture** — return to Camera for a new photo
- **Annotate** — open Landmark Annotation to add/edit markers

CHAPTER 7

Landmark Annotation Tool

Marking vessel reference points for intraoperative orientation.

7.1 Overview

Cyclorotation of the eye during surgery (typically 2°–5°) can shift the effective axis of a toric IOL. Marking pre-operative conjunctival vessel landmarks provides a reliable reference to compensate intraoperatively.

7.2 Adding Landmarks

- 1 Open **Landmark Annotation** from Patient Detail (Annotate button) or from the Alignment Screen quick actions.
- 2 The eye photo is shown full-screen. **Tap any vessel** to place a marker.
- 3 Each marker gets a sequential number and unique colour. Colours cycle through 6 options automatically.
- 4 Use **Undo Last** to remove the most recent marker, or **Clear All** to start over.
- 5 Tap **Save** to store the landmarks in the patient record.

Clinical Tip

Mark 3–4 distinct conjunctival vessels at different positions around the limbus (e.g. 3h, 6h, 9h). Vessels are more reliable than limbal marks as they are fixed relative to the underlying sclera.

7.3 Landmark List

Below the image, a list shows all landmarks with their sequential number, colour, and normalised coordinates (x%, y%) as a percentage of image dimensions. Coordinates are stored resolution-independently.

CHAPTER 8

Alpins Vector Analysis

Quantifying post-operative astigmatism outcomes.

8.1 What is Alpins Analysis?

The **Alpins method** is the internationally recognised standard for reporting astigmatism surgery outcomes. It uses double-angle vector mathematics to quantify the relationship between what was planned (TIA), what was achieved (SIA), and what remains (DV). Access from the **Alpins** button on Patient Detail — TIA is pre-filled from the eye's corneal data.

Pre-op vs Post-op use

For real post-operative analysis, enter the patient's post-op refraction cylinder as the SIA values. The pre-filled SIA (from IOL cylinder) gives a theoretical analysis only.

8.2 Metrics Reference

TIA — Target-Induced Astigmatism

The astigmatic change the surgery was planned to induce. Derived from pre-op corneal data and IOL plan.

SIA — Surgically-Induced Astigmatism

The astigmatic change actually induced. Measured from post-op refraction cylinder.

DV — Difference Vector (residual)

The vector from TIA endpoint to SIA endpoint. Ideal: 0.00 D.

METRIC	IDEAL	GREEN	YELLOW	RED
CI Correction Index	1.00	0.9 – 1.1	0.75 – 1.25	<0.75 or >1.25
ME Magnitude Error	0.00 D	≤ ±0.25 D	> ±0.25 D	—
AE Angle Error	0.0°	≤ ±5°	> ±5°	—
IS Index of Success	0.00	< 0.2	0.2 – 0.5	> 0.5

8.3 CI Interpretation Guide

CI VALUE	INTERPRETATION	IMPLICATION
> 1.0	Overcorrected	IOL cylinder too strong or IOL was over-rotated
= 1.0	Perfect correction	Exact target achieved
< 1.0	Undercorrected	Consider higher cylinder next case; check for axis error

8.4 Double-Angle Plot

The polar plot shows TIA, SIA, and DV vectors in double-angle space (axes doubled to convert 0°–180° TABO to a full circle).

- **Blue dashed** — TIA vector
- **Green dashed** — SIA vector
- **Red solid** — DV (TIA tip to SIA tip)
- Concentric grid at 33%, 67%, and 100% of maximum magnitude

CHAPTER 9

SIA Nomogram Tracker

Building your personalised surgically-induced astigmatism profile.

9.1 Overview

Every surgeon induces a slightly different amount of astigmatism through their incision. The **SIA Nomogram** tracks your personal achieved SIA across cases and calculates your **vector mean**, allowing personalised future calculations. Access from the Patient List → **SIA** button.

9.2 Adding a Case

- 1 Tap **+** **Add Case**.
- 2 Enter **Planned SIA** — magnitude and axis you expected to induce.
- 3 Enter **Achieved SIA** — from post-op refraction.
- 4 Add optional notes (incision type, wound size, etc.).
- 5 Tap **Save**. The mean is recalculated instantly.

9.3 Mean SIA Display

The top of the screen shows your personal **Vector Mean SIA** (mean magnitude, mean axis, case count). Each case is colour-coded by accuracy:

 Green

≤ 0.25 D ·

 Yellow

≤ 0.50 D ·

 Red

> 0.50 D difference.

Recommendation

Build your nomogram with at least

20 cases

before using the mean SIA in routine calculations. Consider separate profiles for different incision locations.

CHAPTER 10

Settings

Configuring the application to your preferences.

10.1 Accessing Settings

Tap the ⚙ gear icon in the top-right of the Patient List screen.

10.2 Available Settings

SETTING	OPTIONS	DESCRIPTION
Default SIA	0.00 – 1.00 D	Pre-populates the SIA field in new eye records
Default SIA Axis	0° – 180°	Pre-populates the SIA axis for your typical incision location
PCA Correction	On / Off	Global default for the PCA toggle in the Toric Calculator
Surgeon Name	Free text	Appears on exported PDF surgical plans
Institution	Free text	Appears on exported PDF surgical plans

CHAPTER 11

Clinical Reference

Quick-reference formulae and axis conventions.

11.1 TABO Axis Convention

DEGREES	CLOCK POSITION	DIRECTION
0° / 180°	3h / 9h	Horizontal
45°	1:30h	Oblique up-right
90°	12h	Vertical (superior)
135°	10:30h	Oblique up-left

WTR (with-the-rule): steep meridian near 90°. **ATR** (against-the-rule): steep meridian near 0°/180°.

11.2 Double-Angle Vector Method (DAVM)

```

J0 = -(C/2) · cos(2 × axis_rad) // WTR/ATR component
J45 = -(C/2) · sin(2 × axis_rad) // Oblique component

// Vector sum of astigmatisms A and B:
Rx = A.C·cos(2·A.axis) + B.C·cos(2·B.axis)
Ry = A.C·sin(2·A.axis) + B.C·sin(2·B.axis)
Result.mag = √(Rx² + Ry²)
Result.axis = ½ · atan2(Ry, Rx) · (180/π) [mod 180°]

```

11.3 Recommended Data Entry Ranges

PARAMETER	TYPICAL RANGE	ALERT RANGE
Corneal astigmatism (K1–K2)	0.25 – 3.00 D	>5.00 D — check for keratoconus

PARAMETER	TYPICAL RANGE	ALERT RANGE
IOL cylinder power	1.00 – 6.00 D	>6.00 D — rarely available
SIA	0.10 – 0.80 D	>1.00 D — verify incision size/type
Target axis	0° – 180°	—

CHAPTER 12

Troubleshooting

Common issues and their solutions.

12.1 Common Issues

PROBLEM	CAUSE	SOLUTION
Camera screen is black	Camera permission not granted	Settings → Privacy → Camera → enable for Zahran Toric Tool
Tilt indicator not responding	Motion sensor unavailable	Restart the app; ensure motion access is granted
Calculator result seems incorrect	K1 entered as steeper than K2	K1 = flat meridian (lower D). Swap K1/K2 values.
Alpins CI shows extreme value (>2.0)	SIA left as IOL cylinder default	Enter actual post-op refraction cylinder for valid analysis
Landmark markers missing on alignment	Landmarks not saved	In Landmark Annotation, verify you tapped Save before closing
PDF export fails	No share sheet available	Restart app; ensure Files or email app is installed
PCA toggle not visible	K values not entered in eye record	Fill K1 Power, K1 Axis, and K2 Power in the eye record first
Patient data not persisting	Storage issue	Check available storage space on device

12.2 Data & Privacy

All patient data is stored **locally on your device only**. No data is transmitted to any server or third party. The app does not require an internet connection. Uninstalling the app will permanently delete all stored patient data — export records as PDFs before uninstalling.

Data Backup

The Zahran Toric Tool does not currently include a cloud backup feature. Export surgical plans as PDFs for important cases to maintain a record outside the app.

12.3 Contact & Support

For questions, feature requests, or bug reports, contact the development team through the app store listing or your institution's IT support channel.

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